

1 Solution — GIAM 4.3

Problem 9

1.1 Problem

Prove the set-theoretic versions of De Morgan's laws using the technique of the previous problems, that is, by double containment.

$$\overline{A \cup B} = \overline{A} \cap \overline{B}, \quad \overline{A \cap B} = \overline{A} \cup \overline{B}.$$

1.2 Answer

Four containments: two per law, each by element-chasing.

Both laws and all four containments were verified over the 1024 pairs of subsets of a five-element universe.

1.3 Law 1: $\overline{A \cup B} = \overline{A} \cap \overline{B}$

Containment $\subseteq$. Let x be an arbitrary element of $\overline{A \cup B}$, so $x \notin A \cup B$.

If x were in A , then x would be in $A \cup B$, contrary to assumption; hence $x \notin A$, that is $x \in \overline{A}$. The identical argument with B in place of A gives $x \in \overline{B}$. Therefore $x \in \overline{A} \cap \overline{B}$.

Containment $\supseteq$. Let x be an arbitrary element of $\overline{A} \cap \overline{B}$, so $x \notin A$ and $x \notin B$.

Suppose, for contradiction, that $x \in A \cup B$. Then $x \in A$ or $x \in B$. The first contradicts $x \notin A$ and the second contradicts $x \notin B$, so both cases are impossible. Hence $x \notin A \cup B$, that is $x \in \overline{A \cup B}$.

Both containments hold, so the sets are equal. ■

1.4 Law 2: $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Containment $\subseteq$. Let x be an arbitrary element of $\overline{A \cap B}$, so $x \notin A \cap B$. Either $x \in A$ or $x \notin A$; these cases are exhaustive by the excluded middle.

Case 1: $x \notin A$. Then $x \in \overline{A}$, hence $x \in \overline{A} \cup \overline{B}$.

Case 2: $x \in A$. If x were also in B then x would lie in $A \cap B$, contrary to assumption. So $x \notin B$, giving $x \in \overline{B}$ and hence $x \in \overline{A} \cup \overline{B}$.

Containment $\supseteq$. Let x be an arbitrary element of $\overline{A} \cup \overline{B}$, so $x \in \overline{A}$ or $x \in \overline{B}$.

Case 1: $x \in \overline{A}$. Then $x \notin A$, so x cannot lie in $A \cap B$, which requires membership in both.

Case 2: $x \in \overline{B}$. Then $x \notin B$, and the same reasoning applies.

Either way $x \notin A \cap B$, that is $x \in \overline{A \cap B}$. ■

1.5 The Picture

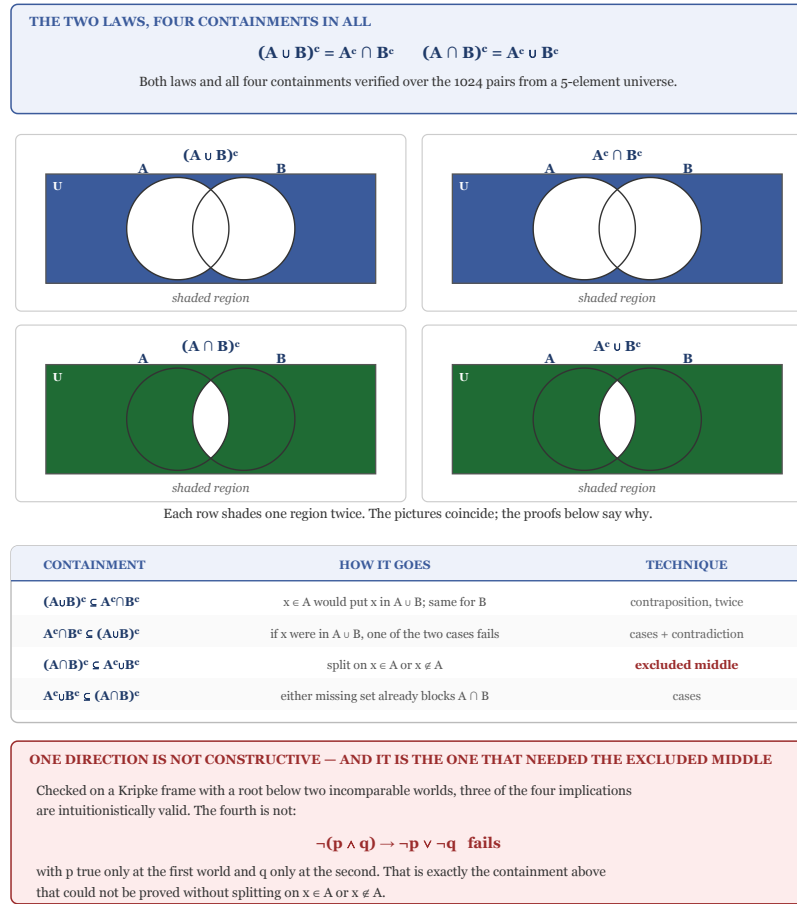


Figure 1.1: A diagram in four parts. The first states the two laws, that the complement of A union B equals the intersection of the complements and the complement of A intersect B equals the union of the complements, and notes that both laws and all four containments were verified over the 1024 pairs from a five-element universe. The second part shows four Venn panels arranged in two rows. The top row, in blue, shades the region outside both circles, first labelled the complement of A union B and then labelled the intersection of the complements; the two pictures are identical. The bottom row, in green, shades everything except the lens where the circles overlap, first labelled the complement of A intersect B and then labelled the union of the complements; again the two pictures are identical. A line notes that each row shades one region twice, that the pictures coincide, and that the proofs say why. The third part is a table of the four containments with the technique each needs. The complement of A union B inside the intersection of complements uses contraposition twice, since x in A would put x in A union B . The intersection of complements inside the complement of A union B uses cases plus contradiction, since if x were in A union B one of the two cases fails. The complement of A intersect B inside the union of complements splits on whether x is in A or not, using the excluded middle, and is flagged in red. The union of complements inside the complement of A intersect B uses cases,

since either missing set already blocks A intersect B . A red band explains that one direction is not constructive, and that it is the one that needed the excluded middle: checked on a Kripke frame with a root below two incomparable worlds, three of the four implications are intuitionistically valid, but the fourth, that not p and q implies not p or not q , fails, with p true only at the first world and q only at the second. That is exactly the containment above that could not be proved without splitting on whether x is in A .

1.6 Remark — Four Containments, Four Different Techniques

Although the two laws look like mirror images, their four halves call on noticeably different logical moves:

$$\begin{array}{l|l} \overline{A \cup B} \subseteq \overline{A} \cap \overline{B} & \text{contraposition, twice} \\ \overline{A} \cap \overline{B} \subseteq \overline{A \cup B} & \text{proof by cases, inside a contradiction} \\ \overline{A \cap B} \subseteq \overline{A} \cup \overline{B} & \text{the excluded middle} \\ \overline{A} \cup \overline{B} \subseteq \overline{A \cap B} & \text{proof by cases} \end{array}$$

The pattern is the one met in 4.3 Problems 7 and 8. A union in the hypothesis hands over a disjunction and forces a case split; an intersection hands over a conjunction and forces none. Complements invert this, since $x \notin A \cup B$ is a *negated* disjunction — which by De Morgan for logic is a conjunction, and so again needs no split.

Only the third row has nothing to split on: its hypothesis $x \notin A \cap B$ is a negated conjunction, which supplies neither a disjunction to case on nor two facts to use. The proof must manufacture its own dichotomy, and it does so out of thin air with $x \in A$ or $x \notin A$. That this is the odd case out is not a presentational accident, as the final remark shows.

1.7 Remark — Either Law Gives the Other

The two laws are not independent results. Apply Law 1 to the sets $\overline{A}$ and $\overline{B}$:

$$\overline{\overline{A} \cup \overline{B}} = \overline{\overline{A}} \cap \overline{\overline{B}} = A \cap B,$$

using $\overline{\overline{X}} = X$, verified for all **32** subsets. Taking complements of both ends gives $\overline{A} \cup \overline{B} = \overline{A \cap B}$, which is Law 2. The substitution was checked directly over all **1024** pairs.

So proving one law suffices, and the same trick runs in reverse. It is the same phenomenon as the duality of 4.3 Problem 3, but made concrete: instead of an abstract rule saying that swapping $\cap$ and $\cup$ preserves truth, here there is an actual map performing the swap.

1.8 Remark — De Morgan Is Why the Duality Principle Works

Complementation is more than a bijection of $\mathcal{P}(U)$. It **reverses inclusion**:

$$A \subseteq B \iff \overline{B} \subseteq \overline{A},$$

verified for all **1024** pairs. Together with $\overline{\overline{A}} = A$ this makes $X \mapsto \overline{X}$ an order-reversing involution — an *anti-automorphism* of the containment order. It carries $\emptyset$ to U , and De Morgan's laws say precisely that it carries $\cup$ to $\cap$ and back.

That is the whole content of the duality principle invoked in Problems 3, 7 and 8. Given any true identity, apply complementation to both sides, push the bars inward with De Morgan, and rename $\overline{A}$ as A : out comes the dual identity, already proved. The principle is not a meta-mathematical convention but a theorem about a specific map, and this problem is that theorem.

Two small consequences worth recording. Complementation turns the smallest set into the largest, so it exchanges the identity elements found in 4.3 Problems 5 and 6. And it commutes with $\triangle$ only up to a twist — $\overline{A} \triangle \overline{B} = A \triangle B$ rather than $\overline{A \triangle B}$ — which is the algebraic shadow of $\triangle$ being addition in a ring of characteristic 2.

1.9 Remark — One of the Four Is Not Constructively Valid

The table above flagged $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$ as the only containment needing the excluded middle. That was not laziness in the write-up: no proof of it can avoid a non-constructive step.

Translating the four containments into propositional implications and testing them on a Kripke frame — a root world below two incomparable worlds, five possible valuations per atom — gives

$$\begin{array}{l|l}
\neg(p \vee q) \rightarrow \neg p \wedge \neg q & \text{valid} \\
\neg p \wedge \neg q \rightarrow \neg(p \vee q) & \text{valid} \\
\neg p \vee \neg q \rightarrow \neg(p \wedge q) & \text{valid} \\
\neg(p \wedge q) \rightarrow \neg p \vee \neg q & \textbf{fails}
\end{array}$$

The failure is witnessed by making p true only at the first upper world and q true only at the second. At the root, $p \wedge q$ can never become true, so $\neg(p \wedge q)$ holds there; but neither $\neg p$ nor $\neg q$ holds at the root, since each atom becomes true somewhere above it. Hence $\neg p \vee \neg q$ fails.

The reading is worth taking seriously. Knowing that A and B have no common element does not, by itself, tell you *which* of the two a given x escapes — and constructively that information has to come from somewhere. Classical logic supplies it free of charge via the excluded middle, which is exactly the step Case 1 and Case 2 of Law 2's first containment rest on. The three other containments never ask which alternative holds, and they survive intact in intuitionistic mathematics.